

HAT-P-32b

R-0049

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_171122 · created 2026-07-09T06:01:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458077.6124 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.054
Transit depth (Rp/Rs)^2	0.00024 +/- 0.017 [%]
Orbital Inclination (inc)	86.14 +/- 1.76
Airmass coefficient 1 (a1)	24.14 +/- 9.35
Airmass coefficient 2 (a2)	-3.0 +/- 1.39
Scatter in the residuals of the lightcurve fit is	33.81 %
Best Comparison Star	#3 - [52, 137]
Optimal Aperture	0.87
Optimal Annulus	4.75
Transit Duration (day)	0.0268 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.002 $\pm$ 0.054 vs 0.1489 (deviation 2.72 σ)
Duration fit vs published	0.0268 d vs 0.1304 d (deviation 25.9 σ)
Mid-time O-C	-3093.9 min
Depth SNR	0.0
Residual scatter	33.81 %

Light curve

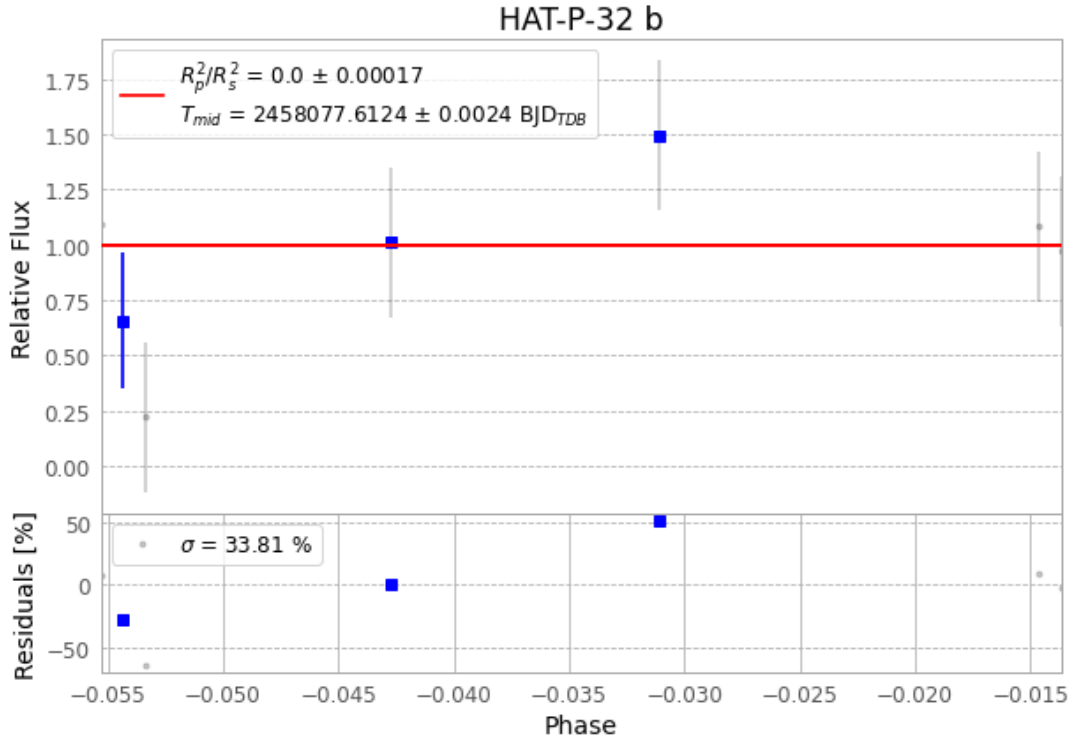


Figure 1 — FinalLightCurve_HAT-P-32 b_2017-11-21.png (SHA-256 f70360f6d59f685d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 323], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 032affec80af646d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

116 FITS frames (76 MB), HATP-32171122031813.FITS → HATP-32171122090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171122.log (3ac453e6e1f7...), FinalLightCurve_HAT-P-32 b_2017-11-21.png (f70360f6d59f...), FinalLightCurve_HAT-P-32 b_2017-11-21.csv (a8dac839f5bf...)

Compute resources

Wall time 162.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.